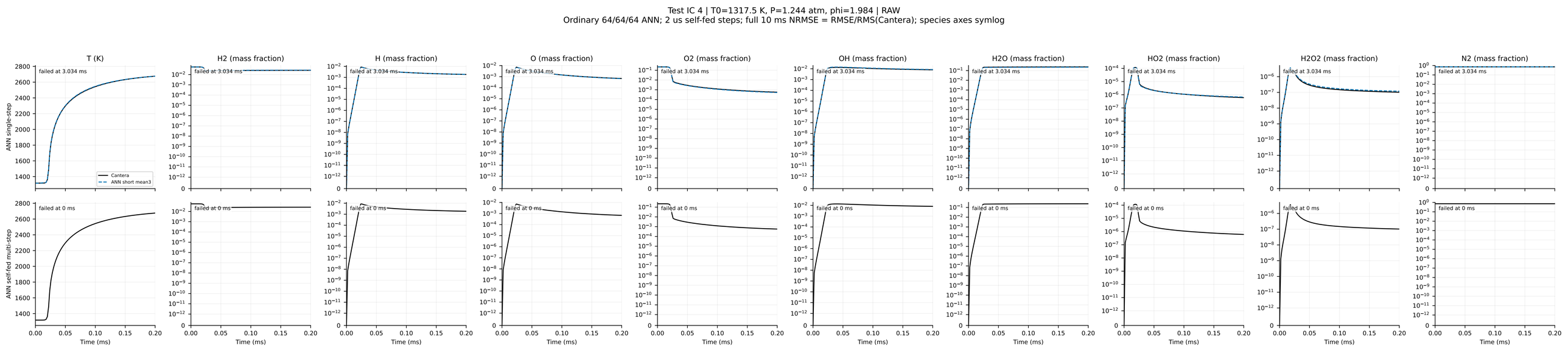
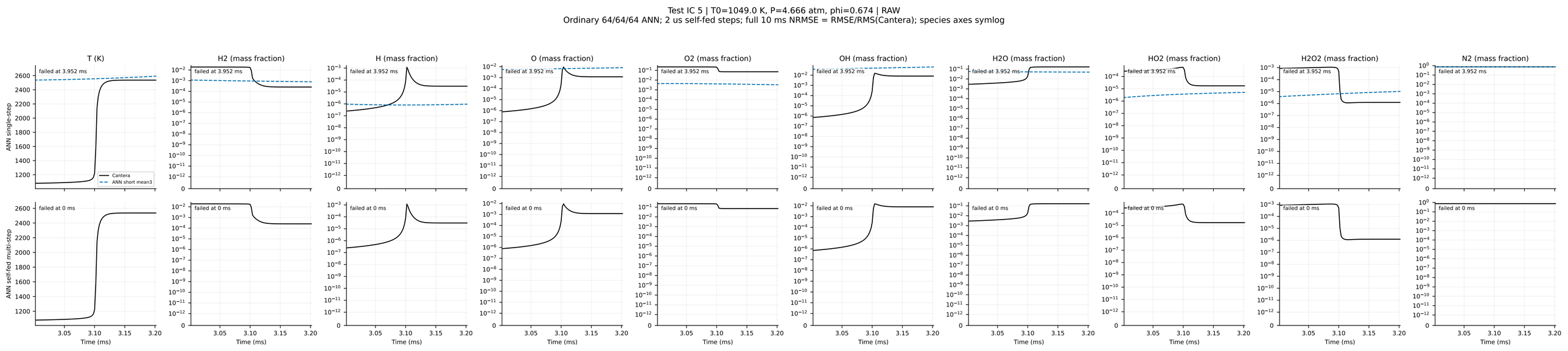
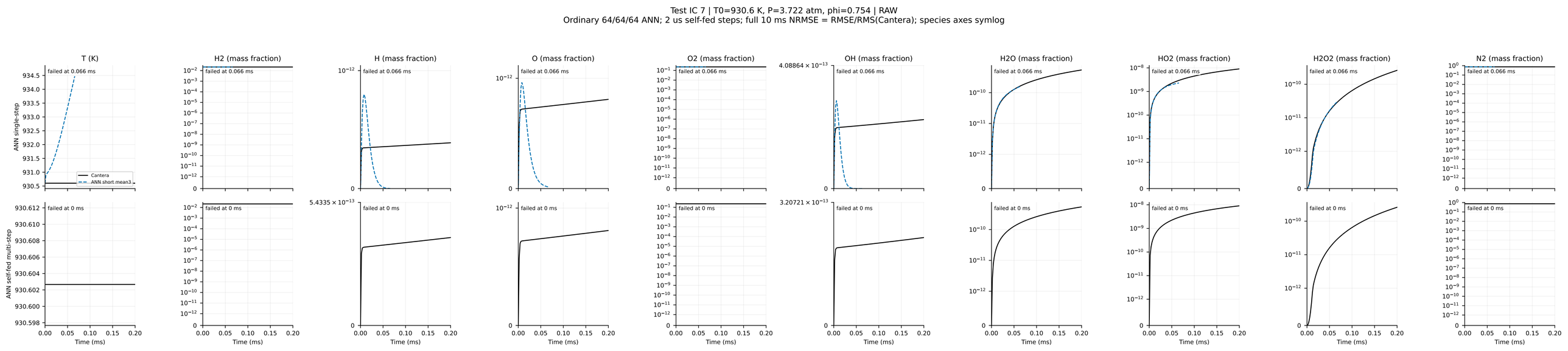
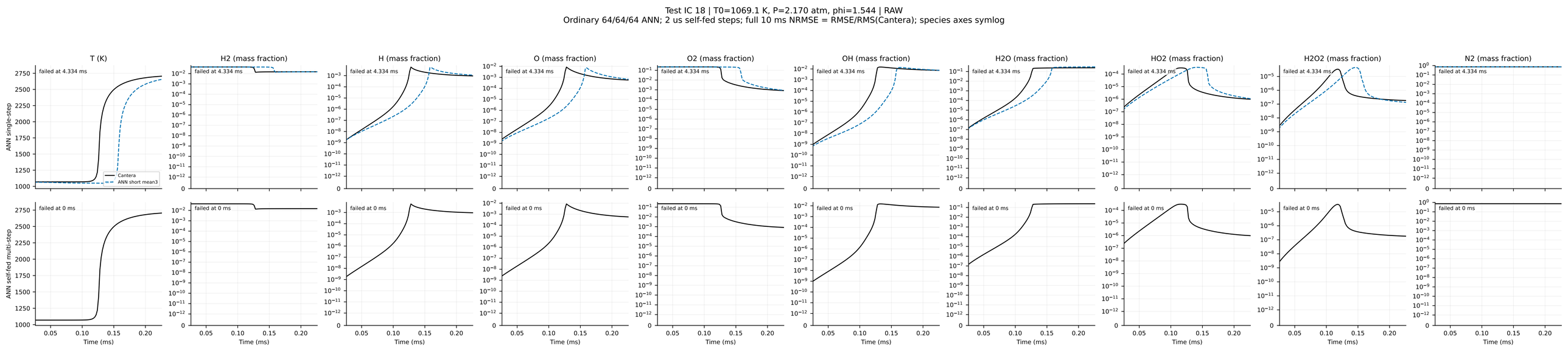
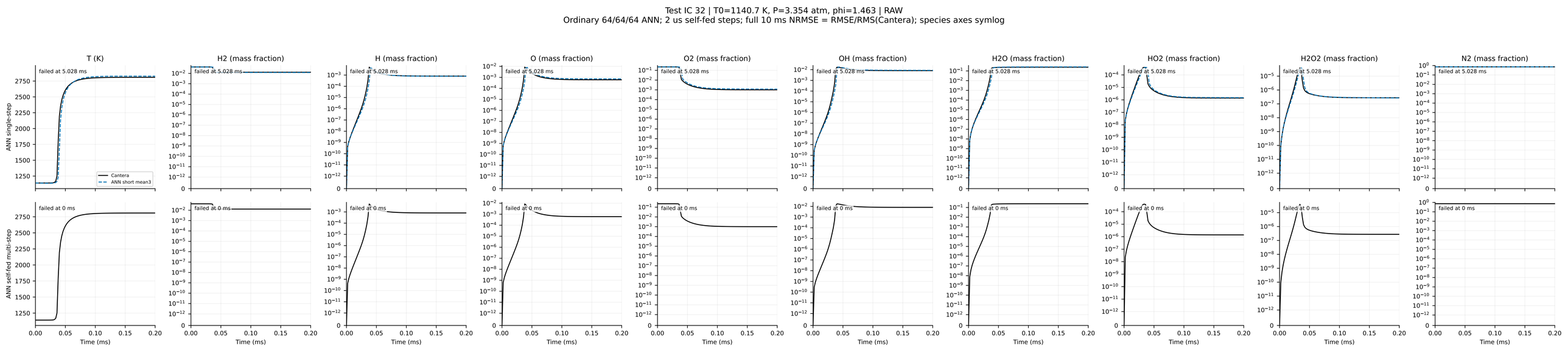
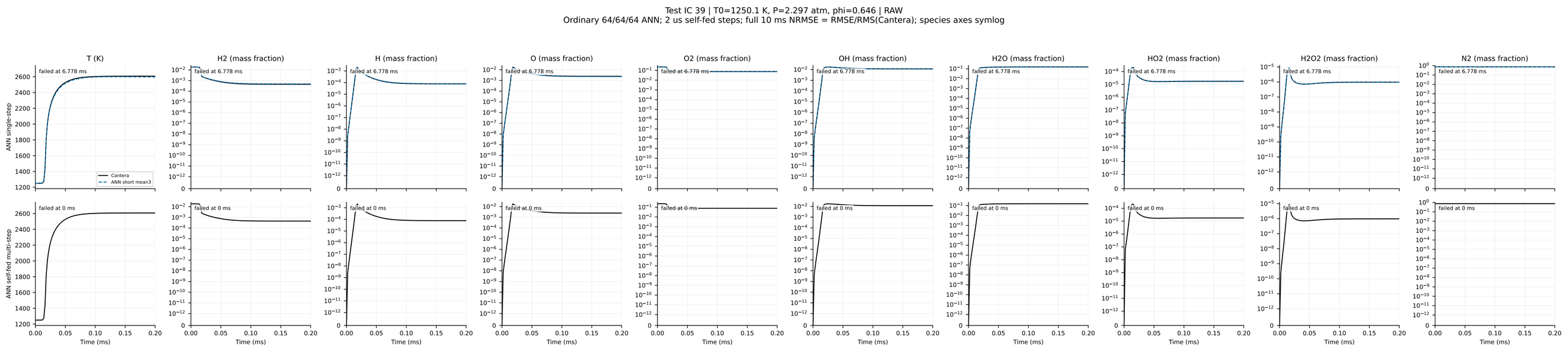

Test IC 4 | T0=1317.5 K, P=1.244 atm, phi=1.984 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
Cantera
ANN short mean3
failed at 3.034 ms
failed at 0 ms
Time (ms)

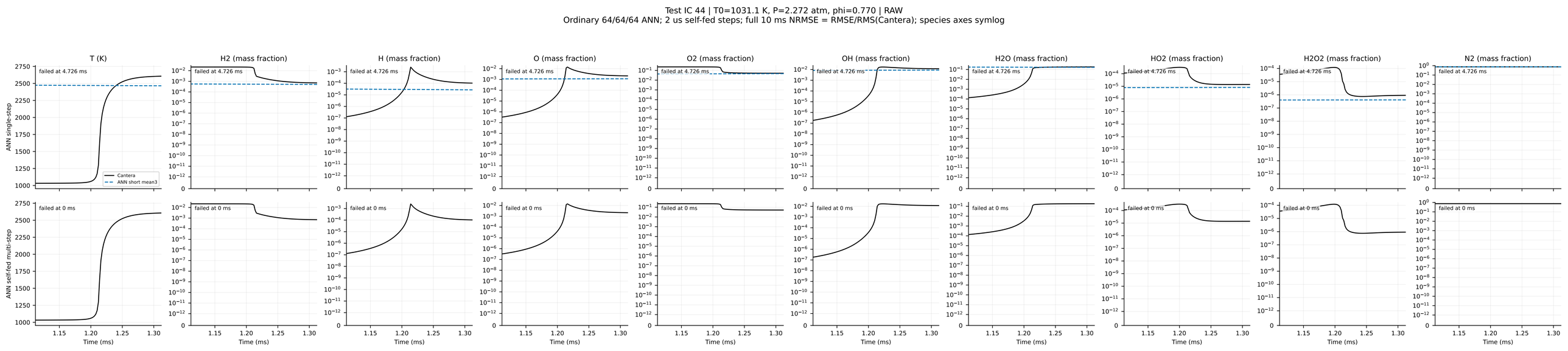

Test IC 5 | T0=1049.0 K, P=4.666 atm, phi=0.674 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 3.952 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)

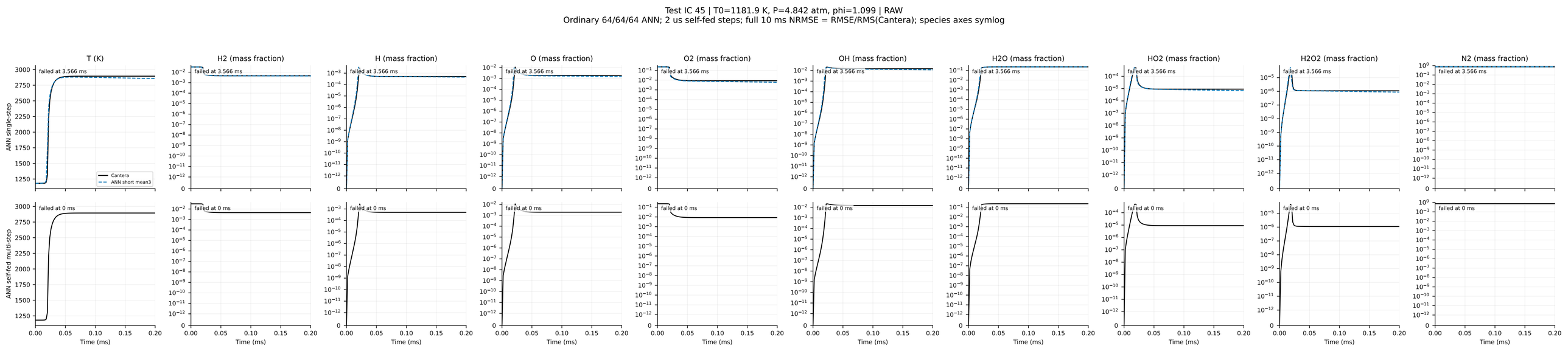

Test IC 7 | T0=930.6 K, P=3.722 atm, phi=0.754 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 0.066 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)

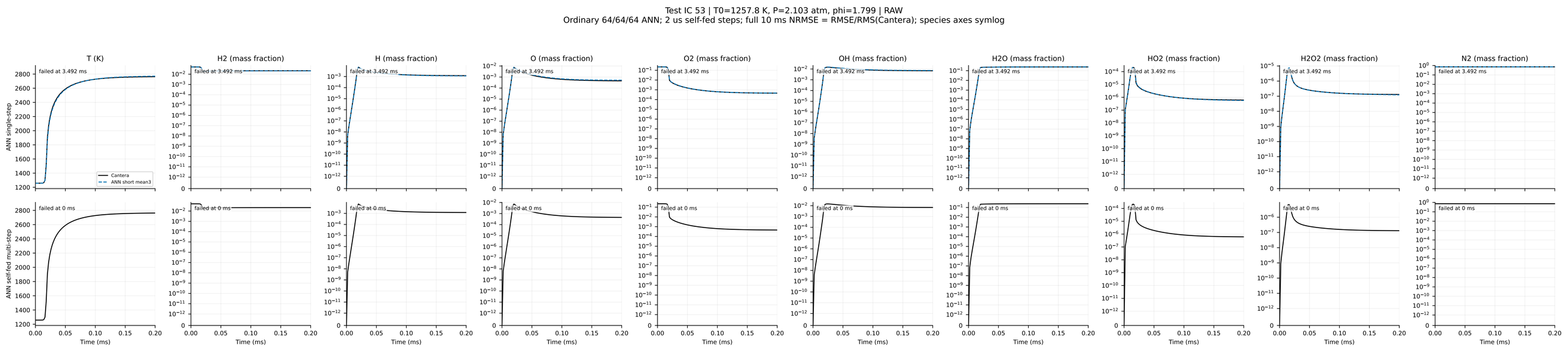

Test IC 18 | T0=1069.1 K, P=2.170 atm, phi=1.544 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 4.334 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)

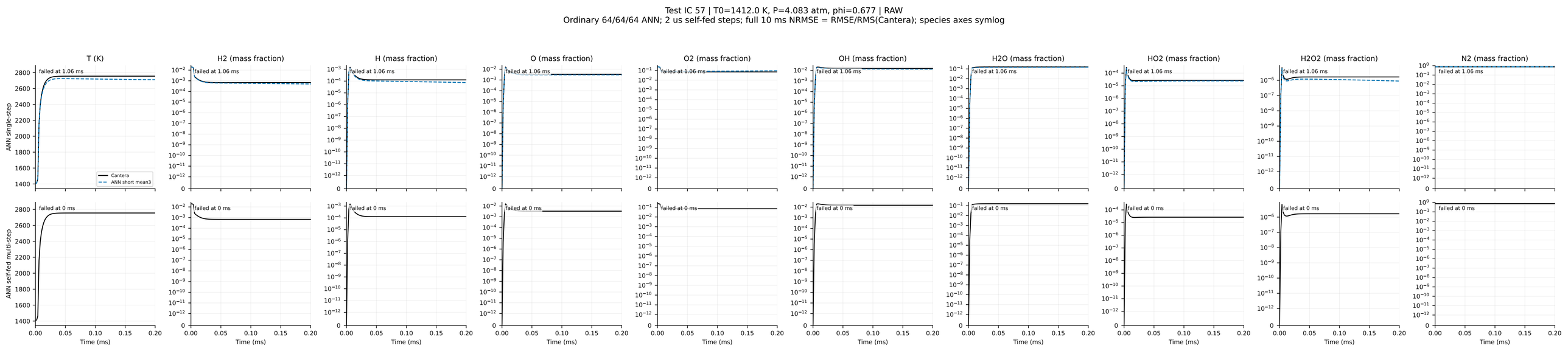

Test IC 32 | T0=1140.7 K, P=3.354 atm, phi=1.463 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 5.028 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)

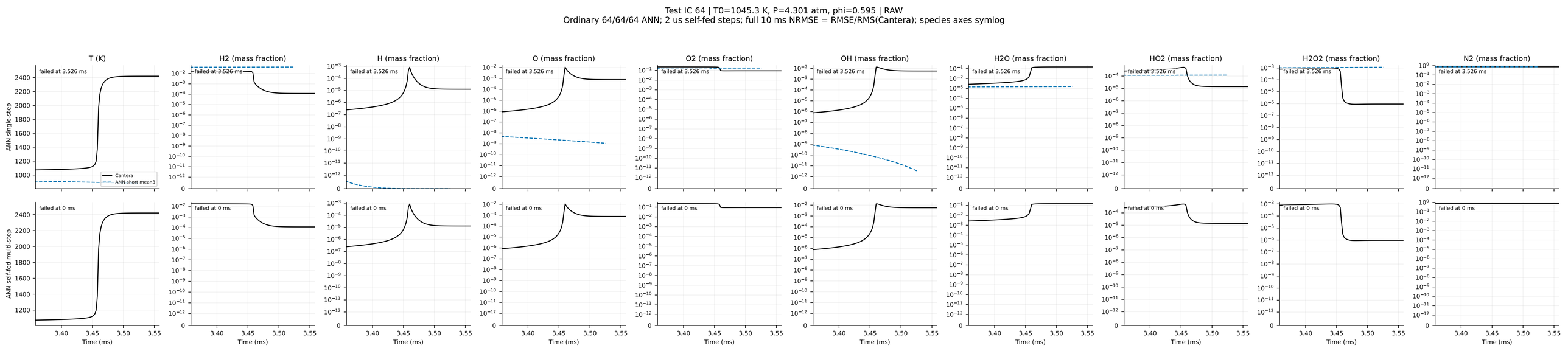

Test IC 39 | T0=1250.1 K, P=2.297 atm, phi=0.646 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 6.778 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)

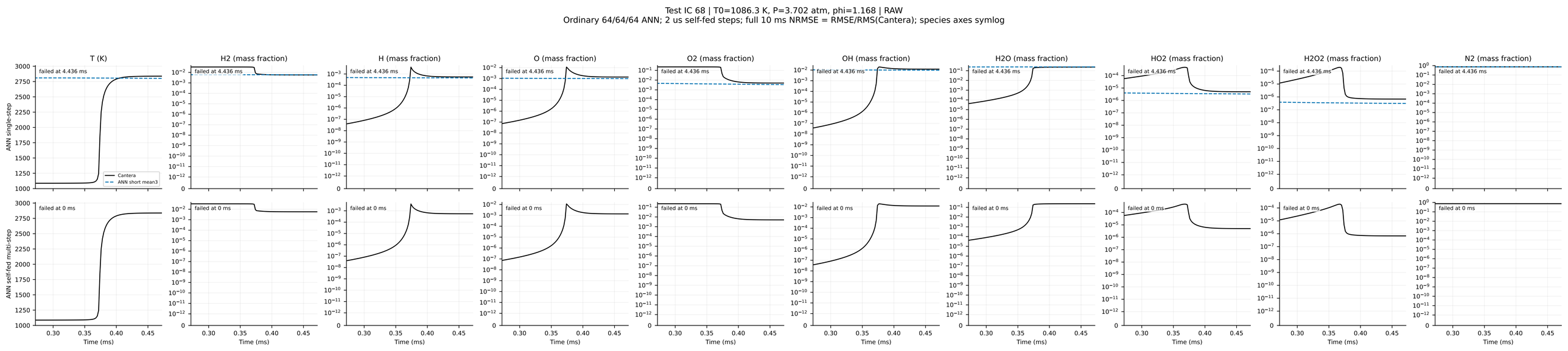

Test IC 44 | T0=1031.1 K, P=2.272 atm, phi=0.770 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 4.726 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)

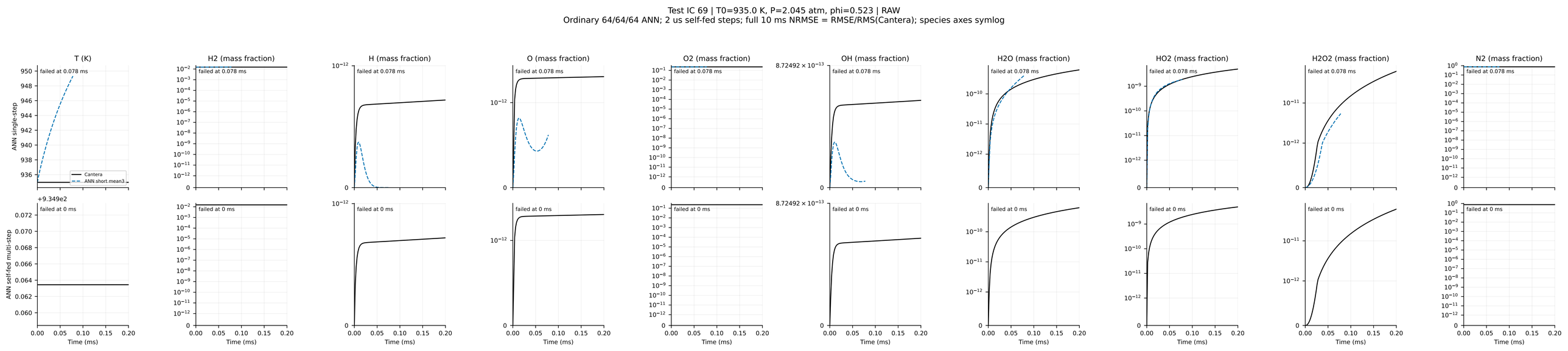

Test IC 45 | T0=1181.9 K, P=4.842 atm, phi=1.099 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 3.566 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)

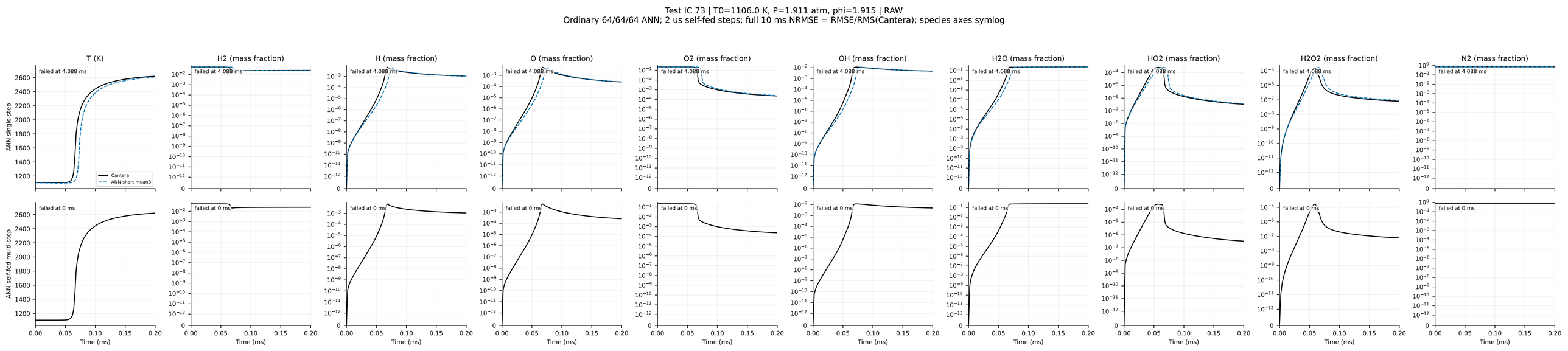

Test IC 53 | T0=1257.8 K, P=2.103 atm, phi=1.799 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 3.492 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)

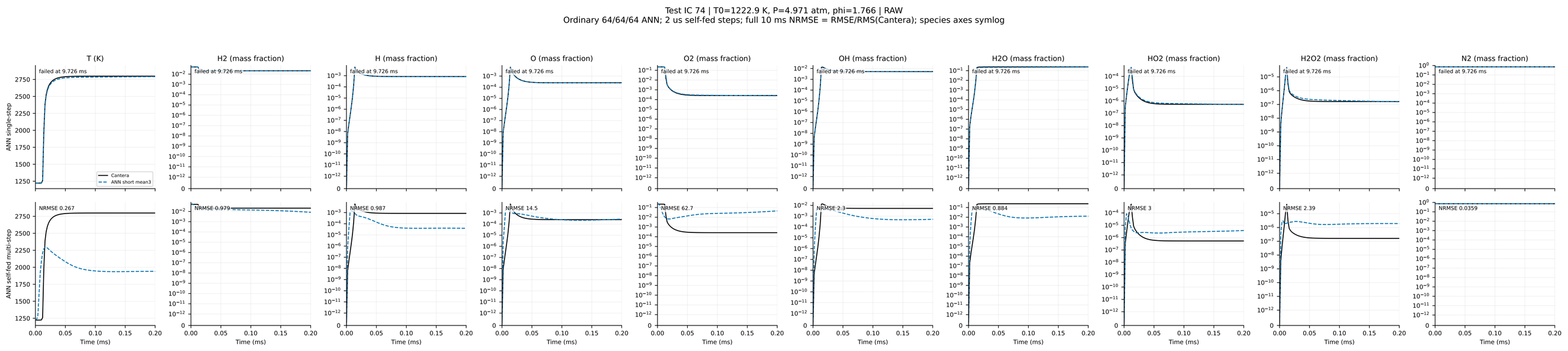

Test IC 57 | T0=1412.0 K, P=4.083 atm, phi=0.677 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 1.06 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)

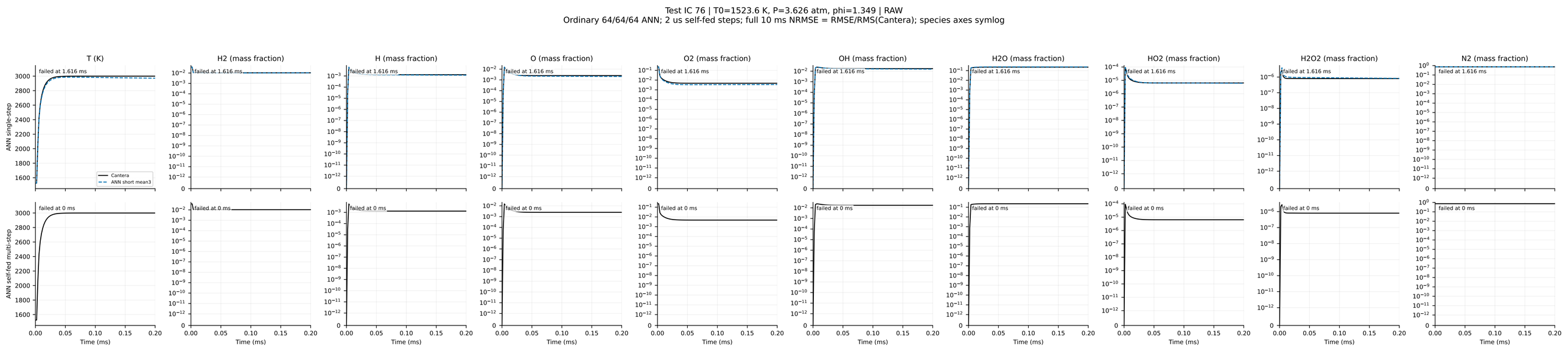

Test IC 64 | T0=1045.3 K, P=4.301 atm, phi=0.595 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 3.526 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)

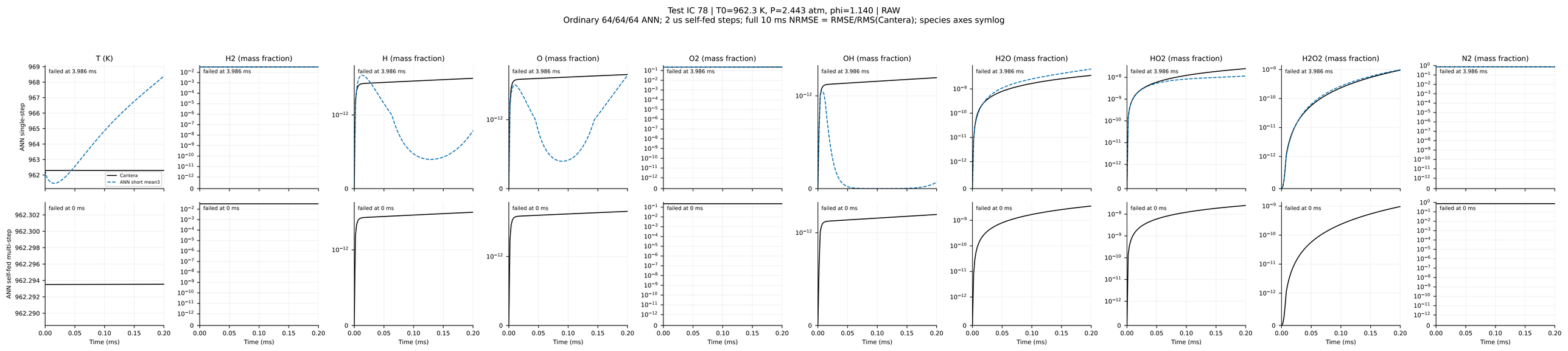

Test IC 68 | T0=1086.3 K, P=3.702 atm, phi=1.168 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 4.436 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)

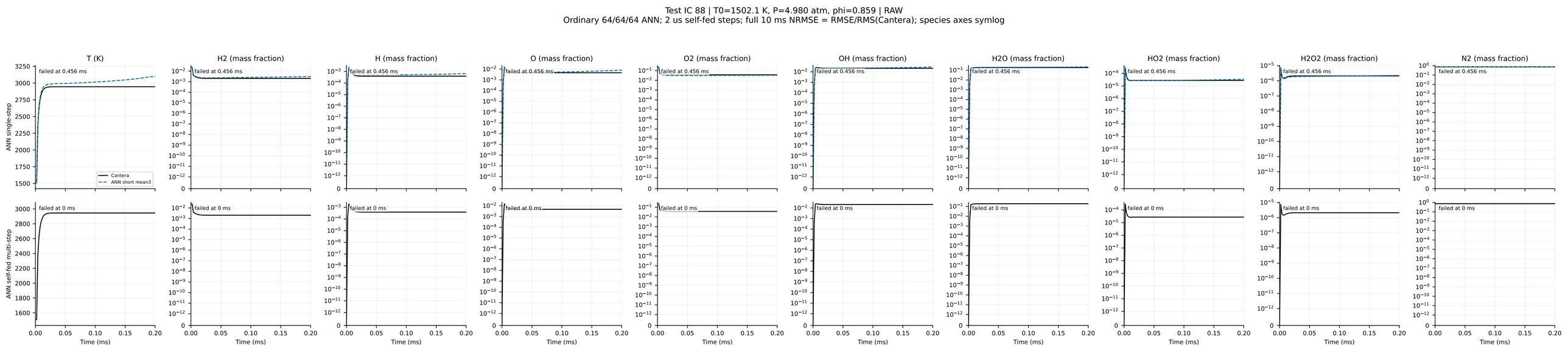

Test IC 69 | T0=935.0 K, P=2.045 atm, phi=0.523 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 0.078 ms
failed at 0 ms
Cantera
ANN short mean3
+9.349e2
8.72492 × 10⁻¹³
Time (ms)


Test IC 73 | T0=1106.0 K, P=1.911 atm, phi=1.915 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 4.088 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)


Test IC 74 | T0=1222.9 K, P=4.971 atm, phi=1.766 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 9.726 ms
Cantera
ANN short mean3
NRMSE 0.267
NRMSE 0.979
NRMSE 0.987
NRMSE 14.5
NRMSE 62.7
NRMSE 2.3
NRMSE 0.884
NRMSE 3
NRMSE 2.39
NRMSE 0.0359
Time (ms)


Test IC 76 | T0=1523.6 K, P=3.626 atm, phi=1.349 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 1.616 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)


Test IC 78 | T0=962.3 K, P=2.443 atm, phi=1.140 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 3.986 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)


Test IC 88 | T0=1502.1 K, P=4.980 atm, phi=0.859 | RAW
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
failed at 0.456 ms
failed at 0 ms
Cantera
ANN short mean3
Time (ms)